

ECON 101—Introduction to Microeconomics
Fall 2023
Midterm I

September 23, 2023

Name:

NU ID number:

V2 Grading Rubric / Solutions

Instructions:

- Answer all the questions. You will be provided scratch paper for your preliminary calculations but your official answers should be written **only in the exam booklet**. **Answers written in scratch paper or outside the space provided will not be considered!**
- The exam has a duration of 2 hours, from 12:00 PM to 2:00 PM.
- The exam has a total of 100 points.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room, but not before 1:45 PM. Otherwise, please remain in the room quietly until the exam concludes.
- Crying is allowed, but please do so quietly to avoid disturbing others.
- Do not use the exam paper to wipe away tears.
- Good luck; you will need it.

Do not write below this area!

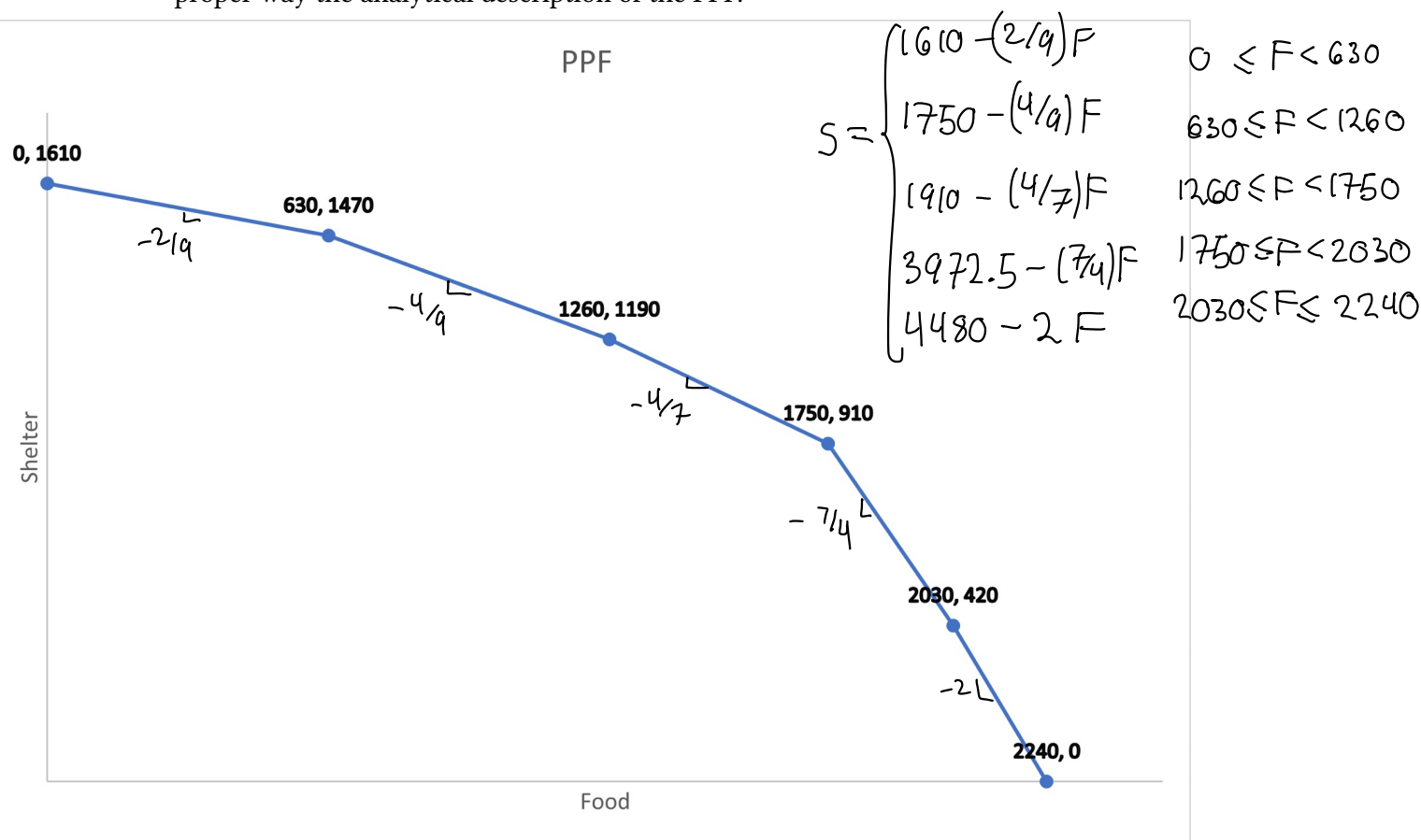
Problem 1	Problem 2	Problem 3	Total

Question 1.

(A)kbilek, (D)aniyar, (M)edina, (S)abina, and (U)lsezim found themselves marooned on the deserted island of NUnesia after their “extreme team-building” cruise took an unexpected detour through the Bermuda Triangle. With their stomachs growling and without any Wi-Fi to pass the time, they soon realized their urgent need to find food and shelter before facing an untimely demise. After a few minutes of contemplation, each of them realized that they could work 10 hours per day for 7 days a week, spending their time gathering food (in kilograms) or building shelter (in square meters). Thinking about their abilities, they concluded that with 1 hour of their time, they would be able to do the following:

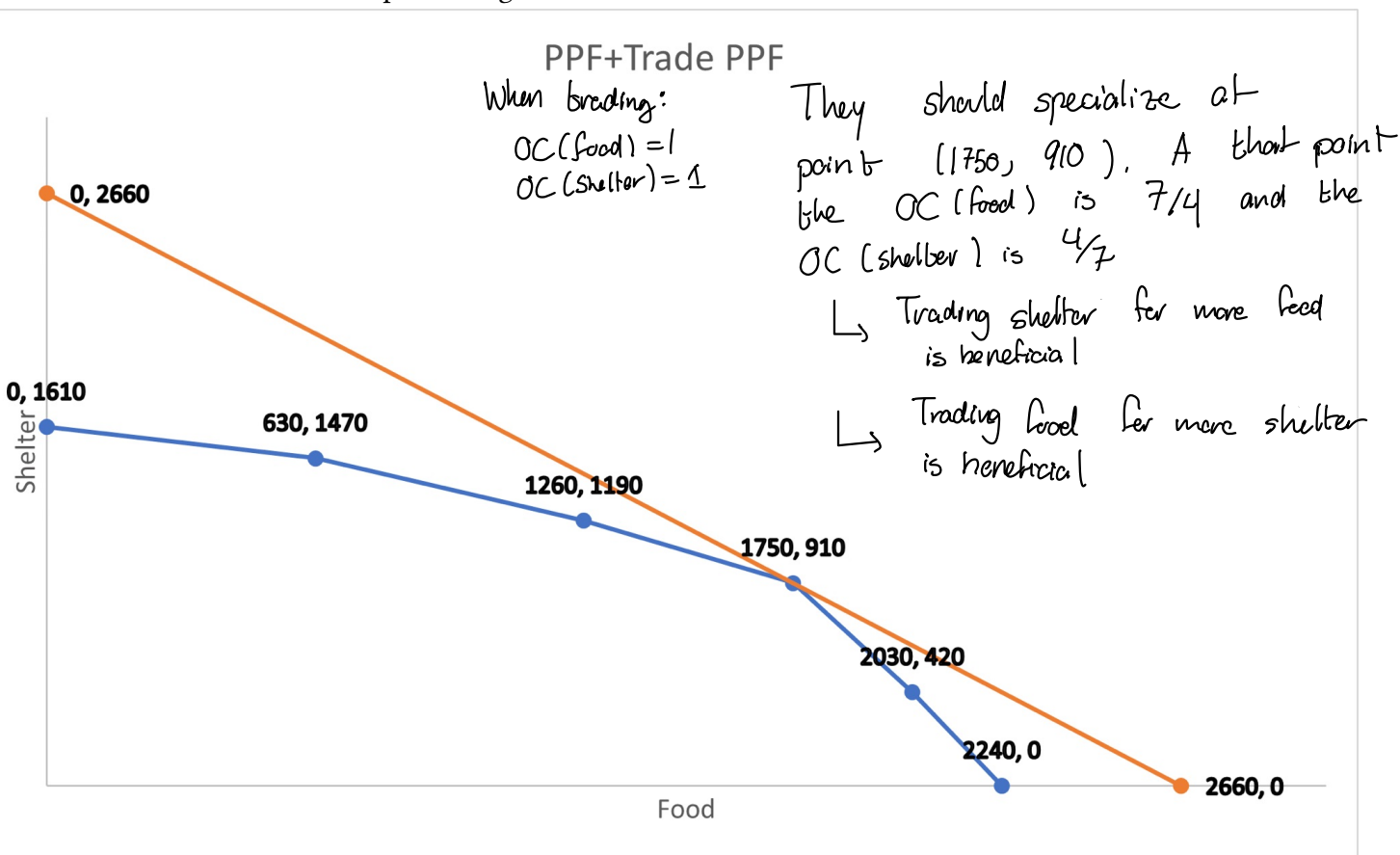
	With 1 hour each can do		O.C. (Food)	Total Food	Total Shelter
	Food	Shelter			
Akbilek	4	7	$7/4$	280	490
Daniyar	3	6	2	210	420
Medina	9	4	$4/9$	630	280
Sabina	7	4	$4/7$	490	280
Ulsezim	9	2	$2/9$	630	140

- (8 P.) If all of them join forces, describe both graphically and analytically their joint production possibilities frontier. Put food on the horizontal axis and shelter in the vertical axis. Be as precise and specific as possible, making sure to label every important point in your diagram, and writing in the proper way the analytical description of the PPF.



Although the group had managed to survive for some time, desperation began to gnaw at our unfortunate castaways. It was in this moment, when all hope had faded, that a glimmer of light emerged on the horizon. Captain Alejandro, aboard the trading ship SS Lepu, sailed toward NUnesia’s shores. Upon disembarking from his ship, he explained that he had come to offer assistance in procuring more supplies. However, he also emphasized the age-old adage that “there’s no such thing as a free lunch”, and thus, they would need to engage in trade for additional supplies. He proposed the following exchange: “I am willing to trade 1 unit of food for 1 unit of shelter.”

2. (6 P.) At the *terms of trade proposed*, at what output combination should our group of castaways specialize so that they can benefit from trading? Redraw the diagram from the last part and show graphically, as well as analytically what the group of castaways should do. Be sure to specify what each of them should be specializing in.



Ulsezim + Medinca + Sabina should specialize in food to produce 1750
 Akbilek + Daniyar should specialize in shelter to produce 910

3. (6 P.) Let the terms of trade be the opportunity cost of obtaining 1 extra unit of food when trading with the Captain. Consider the point of specialization you identified in the previous item. What is the range of terms of trade at which it is still beneficial to specialize in such a point?

Any terms of trade leading to a Opportunity Cost for Food when trading in the interval

$$\left[\frac{4}{7}, \frac{7}{4} \right]$$

Question 2.

In the town of NUburgh, the townsfolk really like chocolate (c) and cupcakes (k). Shynar is an economic consultant for the chocolate and cupcake firms in NUburgh. Shynar and her team have been asked to forecast the economic conditions affecting these markets. In order to do so, she proposes that these markets can be modeled according to the supply and demand equations below:

$$\begin{aligned} Q_c^d &= \alpha_c - \beta_c p_c + \gamma_c p_k & Q_k^d &= \alpha_k - \beta_k p_k + \gamma_k p_c, \\ Q_c^s &= \epsilon_c + \delta_c p_c - w_c L_c & Q_k^s &= \epsilon_k + \delta_k p_k - w_k L_k, \end{aligned}$$

where p_c and p_k are the market prices of chocolate and cupcakes, and all the other variables in Greek letters are parameters of the demand and supply schedules. The terms $w_c L_c$ and $w_k L_k$ are the *total labor costs* for chocolate and cupcake producers respectively, where w_c and w_k is the wage rate per hour for chocolate and cupcake producers, and L_c and L_k are the total labor hours used in chocolate and cupcake production. Total labor costs are determined in the labor markets for each product. Shynar proposes that the labor markets can be described by following demand and supply equations:

$$\begin{aligned} L_c^d &= A_c - B_c w_c & L_k^d &= A_k - B_k w_k, \\ L_c^s &= F_c + G_c w_c & L_k^s &= F_k + G_k w_k, \end{aligned}$$

where the variables with roman letters are parameters.

Akmaral, who is an econometrician in Shynar's team, estimates that the values for the parameters in the model above are according to the table below.

Chocolate Market		Cupcake Market	
Parameter	Value	Parameter	Value
α_c	560	α_k	600
β_c	4	β_k	6
γ_c	2	γ_k	3
ϵ_c	-60	ϵ_k	-30
δ_c	6	δ_k	3
A_c	100	A_k	120
B_c	2	B_k	2
F_c	-20	F_k	-10
G_c	4	G_k	2

Help the team of consultants to figure out what will happen in the chocolate and cupcake markets in NUburgh.

- (5 P.) Compute the labor market equilibria in the chocolate and cupcake markets.

Handwritten: Labor market equilibria solve the following equations

Handwritten: For chocolate mkt.

$$100 - 2w_c^* = -20 + 4w_c^* = L_c^*$$

Handwritten: For cupcake mkt.

$$120 - 2w_k^* = -10 + 2w_k^* = L_k^*$$

Handwritten: Total Labor costs in each mkt.

$$\text{are } TL_c = w_c^* L_c^* ; TL_k = w_k^* L_k^*$$

Handwritten: Given the parameter values above, we have

$$w_c^* = 20 \quad w_k^* = 60$$

$$L_c^* = 60 \quad L_k^* = 50$$

$$TL_c = 1200 \quad TL_k = 3000$$

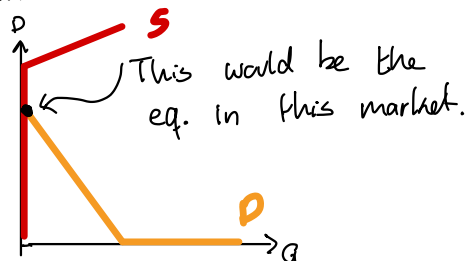
2. (15 P.) Compute the market equilibria for chocolates and cupcakes. Hint: note that the market prices will depend on each other, so they also need to reach an equilibrium.

To find the market equilibrium, we need to solve the following equations:

$$\begin{cases} 560 - 4p_c^* + 2p_h^* = -60 - TL_c^* + 6p_c^* & (C \text{ market}) \\ 600 + 3p_c^* - 6p_h^* = -30 - TL_h^* + 3p_h^* & (H \text{ market}) \end{cases}$$

However, the equations above implicitly assume that (p_c^*, p_h^*) yield positive quantities.

We may instead be facing with a situation like this:



Thus, we must also make sure that at any candidate equilibrium, all quantities are positive. *

We need to make sure that any candidate eq. prices satisfy:

$$\begin{cases} p_c \leq \frac{560 + 2p_h}{4} \\ p_h \leq \frac{600 + 3p_c}{6} \end{cases} \quad \left. \begin{array}{l} \text{maximum willingness to pay} \\ \text{by consumers of each} \\ \text{market} \end{array} \right\}$$

* Remember S and D are not defined for negative quantities!

Solving these two yields the candidates

$$p_c = \frac{21215}{84} \approx 252.56 \quad p_h = \frac{29635}{84} \approx 352.798$$

↳ satisfies the inequality

↳ doesn't satisfy the inequality

↳ yields negative quantity!

Can't be a market equilibrium!

We are in this situation for market K.

$$p_h^* = \frac{600 + 3p_c^*}{6}, \text{ which together}$$

with $560 - 4p_c^* + 2p_h^* = -60 - TL_c^* + 6p_c^*$ yields:

$$p_c^* = \frac{2020}{9}, \quad p_h^* = \frac{1910}{9}$$

$$Q_c^* = \frac{260}{3} \approx 86.67, \quad Q_h^* = 0$$

3. (5 P.) At the market equilibria that you computed above, compute the point-price elasticities of demand and point cross-price elasticities for both chocolates and cupcakes. Are chocolates and cupcakes complements or substitutes?

Own price elasticities

$$\begin{cases} \epsilon_{p_c}^{d,c} = \frac{\Delta \% Q_c}{\Delta \% p_c} = \left(4\right) \frac{p_c^*}{Q_c^*} = \left(-4\right) \left(\frac{2020/9}{260/3}\right) = -4 \cdot \frac{101}{13} \cdot \frac{1}{3} = -\frac{404}{39} \approx -10.36 \\ \epsilon_{p_h}^{d,h} = -\infty \end{cases}$$

Cross-price elasticities

$$\begin{cases} \epsilon_{p_h}^{d,c} = \frac{\Delta \% Q_c}{\Delta \% p_h} = (2) \left(\frac{1910/9}{260/3}\right) = \frac{191}{39} \approx 4.897 \rightarrow \text{complements} \\ \epsilon_{p_c}^{d,h} = \frac{\Delta \% Q_h}{\Delta \% p_c} = \infty \end{cases}$$

4. (15 P.) Suppose that 20 cupcake bakers arrive to NUburch who are willing to work at any prevailing market wage rate. Compute the new equilibria in the labor markets and in the chocolate and cupcakes market.

Labor market equilibria solve the following equations

For chocolate mkt.

$$100 - 2w_c^* = -20 + 4w_c^* = L_c^*$$

For cupcake mkt.

$$120 - 2w_n^* = -10 + 20 + 2w_n^* = L_n^*$$

Total Labor costs in each mkt.

are $TL_c = w_c^* L_c^*$; $TL_n = w_n^* L_n^*$

Given the parameter values above, we have

$$w_c^* = 20 \quad w_n^* = \frac{55}{2}$$

$$L_c^* = 60 \quad L_n^* = 65$$

$$TL_c^* = 1200 \quad TL_n^* = 1787.5$$

This doesn't change anything from the situation in part 2.

Thus: $p_c^* = \frac{2020}{9}$, $p_n^* = \frac{1910}{9}$

$$Q_c^* = \frac{260}{3} \approx 86.67 \quad Q_n^* = 0$$

Question 3.

You may remember the following scenario from one of our past problem sets. There are two villages in which the *individual* markets for honey are described by the following equations:

$$\begin{aligned}q_1^d &= 300 - 2p \\ q_1^s &= -20 + 4p\end{aligned}$$

$$\begin{aligned}q_2^d &= 200 - 3p \\ q_2^s &= 40 + 3p\end{aligned}$$

1. (2 P.) Calculate the equilibrium in each individual honey market.

$$\begin{aligned}q_1^* &= 300 - 2p_1^* = -20 + 4p_1^* \\ p_1^* &= \frac{160}{3}, \quad q_1^* = \frac{580}{3}\end{aligned}$$

$$\begin{aligned}q_2^* &= 200 - 3p_2^* = 40 + 3p_2^* \\ p_2^* &= \frac{80}{3}, \quad q_2^* = 120\end{aligned}$$

2. (4 P.) You may also remember Panqueque, the rising entrepreneur. Panqueque decides to earn some money by taking opportunity of an *arbitrage opportunity*: buying some honey for a lower price in one of the villages and reselling it for a higher price in another village. Suppose that Panqueque decides to buy a quantity of x jars of honey from the village with the lower price and then resell that quantity x to the village with higher price. Show (algebraically) how this decision influences the demand and/or supply of each village. Calculate the new equilibrium prices as a function of x .

$$\begin{aligned}q_1^s &= -20 + x + 4p \\ \text{Solving together with } q_1^d &\text{ \& } q_2^s \text{ yields} \\ p_1^*(x) &= \frac{320 - x}{6} \qquad p_2^*(x) = \frac{160 + x}{6}\end{aligned}$$

3. (4 P.) Assume that Panqueque has no transportation/logistical costs from buying/selling jars of honey from one market to the other. Find the optimal quantity x^* that Panqueque should buy from the village with the lower price and then resold in the village with higher price if he wanted to get the most profit from his fledgling enterprise, and compute the optimal level of profits that Panqueque will obtain.

$$R_{\text{venues}}(x) = x(p_1^*(x) - p_2^*(x))R(x)$$

$$\text{Let } \bar{x} \text{ s.t. } p_1^*(\bar{x}) - p_2^*(\bar{x}) = 0$$

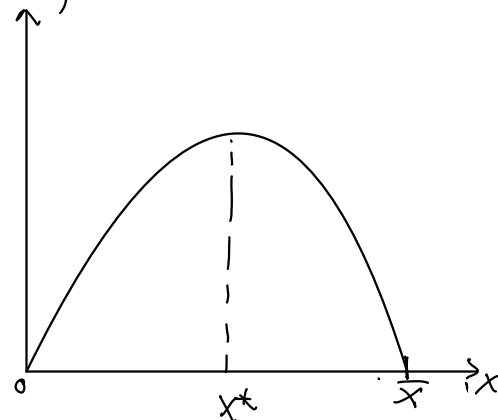
$$\Rightarrow x^* = \bar{x}/2 \quad (\text{Can also be done with calculus})$$

$$\frac{320 - \bar{x}}{6} = \frac{160 + \bar{x}}{6}$$

$$\bar{x} = \frac{320 - 160}{2} = 80$$

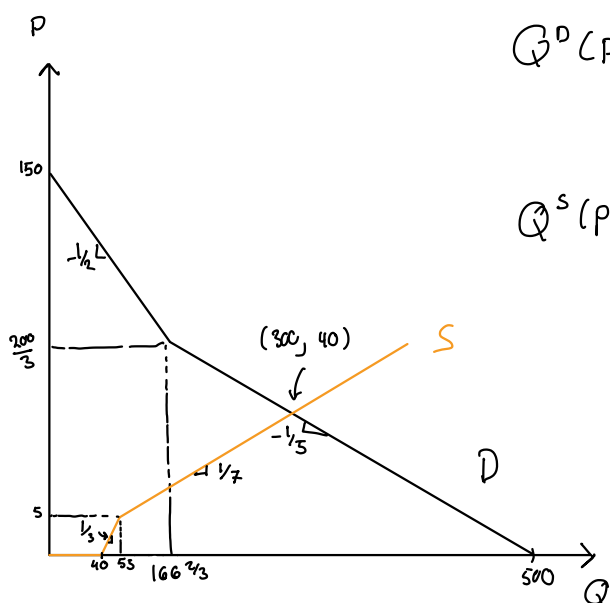
$$\Rightarrow x^* = 40$$

$$R(x^*) = \frac{1600}{3}$$



An NU Economics professor, call him Prof. A, told Panqueque that he should use his deep knowledge of Economics to engage in a more ambitious and profitable enterprise.¹ He recommended that Panqueque puts the infrastructure necessary to *join all* the buyers and sellers in both individual village markets, and that Panqueque should act as an *intermediary* between all buyers and sellers, charging a fee for his services.

4. (5 P.) First of all, Panqueque needs to figure out the aggregate demand and supply for both villages. Help Panqueque by computing the analytical and graphical description of the aggregate demand and supply curves. Be as precise as possible.



$$Q^D(p) = \begin{cases} 300 - 2p & \frac{200}{3} \leq p \leq 150 \\ 500 - 5p & 0 \leq p < \frac{200}{3} \end{cases}$$

$$Q^S(p) = \begin{cases} 40 + 3p & 0 \leq p < 5 \\ 20 + 7p & 5 \leq p \end{cases}$$

¹Prof. A says the same thing to all his Economics students.

Panqueque decides to name his intermediary service as *PanquequeExpress*.² For his intermediary services, Panqueque charges producers a fee F per unit sold. This means that if the per unit price at which honey is sold to consumers is p , then producers will only receive $p - F$. Thus, the fee F not only alters the market equilibrium, but also the price that the producers receive, which will all depend on the fee F .

5. (10 P.) In the joint market, compute the new market equilibrium price $p^e(F)$ and quantity $Q^e(F)$, as well as the effective price received by producers $p^e(F) - F$. All your expressions should be a function of the fee F .

① Assume the mkt. eq. with the fee is in the range of prices $5 \leq p \leq 200/3$, then:

$$20 + 7(p - F) = 500 - 5p$$

$$p^e(F) = \frac{480 + 7F}{12} \quad Q^e(F) = 500 - 5\left(40 + \frac{7}{12}F\right)$$

$$= 40 + \frac{7}{12}F \quad = 300 - \frac{35}{12}F$$

$$p^e(F) - F = 40 - \frac{5}{12}F$$

② Assume the mkt. eq. with the fee is in the range of prices $p > 200/3$, then:

$$20 + 7(p - F) = 300 - 2p$$

$$p^e(F) = \frac{280 + 7F}{9}$$

$$p^e(F) - F = \frac{280 - 2F}{9}$$

$$Q^e(F) = 300 - 2\left(\frac{280 + 7F}{9}\right)$$

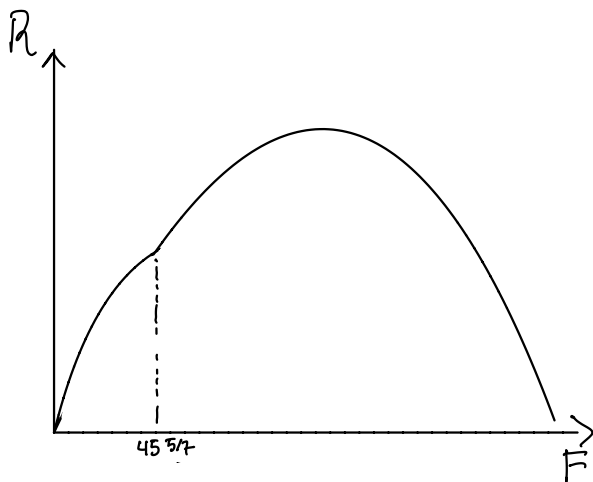
$$= \frac{2700 - 560 - 14F}{9}$$

$$= \frac{2140 - 14F}{9}$$

²No connection with AliExpress.

6. (5 P.) For simplicity, suppose that PanquequeExpress does not incur in any logistical/transportation costs from its intermediary services. Compute the revenue of PanquequeExpress as a function of the fee F : $R(F) = F \times Q^e(F)$.

$$R(F) = \begin{cases} F \left(300 - \frac{35}{12} F \right) & p^e(F) \leq \frac{200}{3} \\ F \left(\frac{2140 - 14F}{9} \right) & p^e(F) > \frac{200}{3} \end{cases} = \begin{cases} F \left(300 - \frac{35}{12} F \right) & 0 \leq F \leq 45 \frac{5}{7} \\ F \left(\frac{2140 - 14F}{9} \right) & F > 45 \frac{5}{7} \end{cases}$$



$$\begin{aligned} p^e(F) &\leq \frac{200}{3} \\ \updownarrow \\ 40 + \frac{7}{6} F &\leq \frac{200}{3} \\ F &\leq 45 \frac{5}{7} \end{aligned}$$

7. (10 P.) If PanquequeExpress seeks to maximize revenues, what is the optimal fee F^* ? At the optimal fee, what are the revenues of PanquequeExpress?

From the previous, you can see that optimal revenues should be computed from $R(F)$ when $F > 45 \frac{5}{7}$

$$\Rightarrow F^* = \frac{2140/14}{2} = 76.43$$

$$R(F^*) = 9086.5 //$$